THE FUTURE IS NOW

HOLOGRAMMED WORK MEETS, SKIN PATCHES TO MONITOR HEALTH 24X7, EMBEDDED DEVICES THAT PROMISE A SENSORY OVERLOAD...6G WILL REDEFINE HOW WE WORK, LIVE AND PLAY



In communications technology, the focus of a network changes with each generation. If the 2G and 3G eras were about human-to-human communication through voice and text, 4G heralded a fundamental shift to the consumption of data, and 5G concentrated on tying up the Internet of Things (IoT) and industrial automation systems. In the world of 6G, the digital, physical and human will fuse seamlessly to trigger extrasensory experiences. Intelligent knowledge systems will combine with computation capabilities to make humans endlessly efficient and redefine how we live, work and play.

WHY IT IS A GAME CHANGER

For sure, 6G will be a 'sixth sense' experience for humans where physical sense will meet Artificial Intelligence (AI). How exactly? Start with the basic enabler: vastly higher frequencies than in 5G. That has two key effects. One, substantially lower latency—the time taken by data to travel from a source to the end user. And two, higher bandwidth—which determines the amount of data that can travel over the network at a time. 6G is said to support 1 microsecond-latency communication, which even at a rough count yields communication and data transfer speeds several hundred times greater than what you have now.

What will it mean for you? Expect to be pampered by vast improvements in things like touch control, imaging and location awareness, particularly when 6G partners with AI. So self-driving cars will be able to communicate with each other, navigating pedestrians as well as traffic. Smartphones will likely be replaced by smart wearable devices, headsets and implants that will accessorise human thought and action in real time. Expect augmented reality to replace virtual reality and holographic technology to be integrated into many apps. So, say, instead of a video conference, you could have real-time conversations with people in virtual reality, with wearable sensors giving you the feel of being in the same room together.

WHAT INDIA NEEDS TO MASTER

India has recently secured 127 global patents on 6G and, if IT minister Ashwini Vaishnaw is to be believed, is aspiring for a 10 per cent share in patents by 2030. What is critical is to develop unified human-machine and



"Wireless technologies will be one of India's biggest tech exports in the future. Driving industry adoption and transforming enterprises to a 'digital first' operation is key for this"

–SHAMIK MISHRA
V-P and CTO, Connectivity, Capgemini Engineering